uranium still exist in the world, and that these actually decay only very slowly. In fact, their atoms have existed in an unchanged form for several thousand millions of years. The same applies to thorium and actinium. As a matter of fact, if these three long-lived radioactive elements did not exist, radioactivity would have disappeared from the world a long time ago, except for a few of the lighter elements. For the majority of the other natural radioactive substances originate from these, and are much shorter-lived. The products of the decay of a substance are designated collectively as a *radioactive series.* There are three natural radioactive series: the *uranium series,* the *thorium series,* and the *actinium series.*

It was stated in the third lecture that the *half-life* is used as the measure of the life span of a radioactive substance; the half-life is the period of time during which one-half of the number of atoms present at the start of the period disintegrates. The half-life periods of the various alpha emitters show extraordinary differences in order of magnitude. The half-life of uranium, for instance, is 4,600 million years, whereas the half-life of one of its daughter elements, radium C', is just about one one-millionth of a second. Between these two extremes there are all conceivable intermediate values; for instance, the half-life of radium is 1,580 years. This raises the question as to the cause of these vast differences.

In this connection, it is a very important fact that there exists a simple relationship between the energy of the alpha particles and the half-life of the substance in question. This relationship was discovered, relatively early, by Geiger and Nuttall. These two scientists found that, on the whole, there exists a linear relationship between the logarithm of the decay probability (the reciprocal of the average life of the atom) and the energy of the alpha particles. The latter, as indicated by their equal range (Figure 3), is a magnitude characteristic of every specific radioactive substance. The greater is the quantity of energy available for the decay process, and the greater is the energy of the particle, the more quickly will the decay occur, on the average. If λ designates the decay probability, discussed in the second chapter of this book, and E the energy of the alpha particle, we can write the Geiger-Nuttall relationship in the following form:

$$\log\lambda = A + BE$$

where A and B are constants, to be determined experimentally. The value of the decay coefficient can be obtained from the already mentioned equation N